Proof Systems for a Gödel Modal Logic

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Abstract. A basic propositional modal fuzzy logic $\mathsf{GK}_\Box$ is defined by combining the Kripke semantics of the modal logic K with the many-valued semantics of Gödel logic G. A sequent of relations calculus is introduced for $\mathsf{GK}_\Box$ and a constructive counter-model completeness proof is given. This calculus is used to establish completeness for a Hilbert-style axiomatization and Gentzen-style hypersequent calculus admitting cut-elimination, and to show that the logic is PSPACE-complete.

1 Introduction

Logical formalizations of vagueness and modal notions such as necessity, knowledge, and obligation have been studied intensively, the former primarily as *fuzzy logics* (see e.g. [14,16]), the latter under the rubric of *modal logics* (see e.g. [7]). However, there have been few treatments and until recently no systematic study of *modal fuzzy logics*. Such an investigation is important for providing a unified approach for topics such as fuzzy description logics, which can be understood, analogously to classical description logics, as multi-modal fuzzy logics [19,15].

Particular examples of fuzzy modal logics given in the literature have typically been situated quite far up the spectrum of modal logics, e.g. at the level of the logic $\mathsf{S5}$ (see e.g. [14]) or focus just on the minimal fuzzy logic of Zadeh (see e.g. Zhang [21]). More general approaches dealing with many-valued modal logics, such as [11,12], have focussed on the finite-valued case. Recent papers of Priest [17] and Bou et al. [5] provide a broad basis for studying fuzzy modal logics but again the majority of the results concern finite-valued modal logics. Deferring formal definitions to the next section, the rough idea of these approaches (followed also in this paper) is to consider Kripke models where the accessibility relation between worlds may be either Boolean-valued or many-valued. Propositional connectives operate as usual for the logic in question at an individual world, while the values of boxed formulas $\Box A$ are calculated using the infimum of values of A at accessible (to some degree) worlds. Validity is defined as usual as truth (i.e. taking the value 1) at all worlds of all models.

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In this paper, we narrow our focus on the fuzzy side to Gödel logic G, the infinite-valued version of a family of finite-valued logics introduced by Gödel in the 1930s [13], axiomatized by Dummett by adding the schema $(A \to B) \vee (B \to A)$ to intuitionistic logic [9]. Aside from being an important fuzzy logic, there is also a good practical reason to focus on G in modal contexts. As noted in [5], G is the only fuzzy logic whose modal analogues admit the schema $\Box(A \to B) \to (\Box A \to \Box B)$ (roughly speaking, since G, unlike other fuzzy logics, admits both weakening and contraction). Moreover, axiomatizations for a basic "K" Gödel modal logic with the $\Box$ or the $\Diamond$ modality (so far not both) have recently been provided by Caicedo and Rodríguez [6]. These authors also show, interestingly, that the logic with $\Box$ is complete with respect to either Boolean or many-valued accessibility relations but does not have the finite model property, while the logic with $\Diamond$ based on a many-valued accessibility relation has the finite model property but differs from the corresponding logic with a Boolean-valued relation.

We extend the work of [6] here by providing proof systems for the Gödel modal logic with $\Box$, the broader aim being to initiate a general investigation into the proof theory of modal fuzzy logics (e.g., as undertaken for fuzzy logics in [16]). A wide range of proof calculi have been developed for Gödel logic, including the sequent calculi of Sonobe [18] and Dyckhoff [10]; here we focus on extending the hypersequent calculus of Avron [2] and sequent of relations calculus of Baaz and Fermüller [4]. In particular, we give a constructive proof of completeness for the more proof-search-oriented sequent of relations calculus and use it to give both an alternative completeness proof for the axiomatization of Caicedo and Rodríguez [6] and a (first) proof of PSPACE-completeness. We then establish completeness and cut-elimination for the more elegant extended hypersequent calculus, better suited e.g. for extension to first-order modal fuzzy logics or investigating theoretical properties such as interpolation.

Note that the approach taken both here and in [6] for Gödel logic differs markedly from certain other developments in the literature. In particular, the intermediate logics extended with modalities investigated in e.g. [20] make use of two accessibility relations in Kripke models, one for the modal operator and another for the intuitionistic connectives. Also, the class of modalities for fuzzy logics considered in [8] represent truth stressers such as "very true" and, unlike the modalities considered here, can be interpreted as unary functions on $[0, 1]$.

2 The Gödel Modal Logic $\mathsf{GK}_\Box$

We make use of a language $\mathcal{L}_\Box$ with a countably infinite set Var of variables $p, q, r \ldots$, binary connectives $\to$, $\wedge$, $\vee$, constants $\bot$, $\top$, and a unary connective $\Box$. The set Fm of formulas, denoted $A, B, C \ldots$, is defined inductively as usual, and the complexity of a formula A, denoted $|A|$, is the number of connectives occurring in A. We also let $\neg A =_{\text{def}} A \to \bot$ and $A \leftrightarrow B =_{\text{def}} (A \to B) \wedge (B \to A)$. We use $\Gamma, \Pi, \Sigma, \Delta$ to stand for finite multisets of formulas, writing $\bigvee \Gamma$ and $\bigwedge \Gamma$ with $\bigvee [] =_{\text{def}} \bot$ and $\bigwedge [] =_{\text{def}} \top$ for disjunctions and conjunctions of formulas and $\Box\Gamma$ for $[\Box A : A \in \Gamma]$. We let $\Gamma^0 =_{\text{def}} []$ and $\Gamma^{n+1} =_{\text{def}} \Gamma \uplus \Gamma^n$ for $n \in \mathbb{N}$.

Following similar ideas in [14,6,5], the usual definition of the modal logic K can be generalized to define a natural "Gödel modal logic" $\mathsf{GK}_\Box$. Recall that a *(standard) Kripke frame* is a pair $\langle W, R\rangle$ where W is a non-empty set of *worlds* and $R \subseteq W^2$ is a binary *accessibility relation* on W. A *Kripke model for* $\mathsf{GK}_\Box$ is then a 3-tuple $K = \langle W, R, V\rangle$ where $\langle W, R\rangle$ is a Kripke frame and $V : Var \times W \to [0,1]$ is a mapping, called a *valuation*, extended to $V : Fm \times W \to [0,1]$ by:

- $V(A \to B, w) = \begin{cases} V(B,w) & \text{if } V(A,w) > V(B,w) \\ 1 & \text{otherwise;} \end{cases}$
- $V(A \wedge B, w) = \min(V(A,w), V(B,w))$;
- $V(A \vee B, w) = \max(V(A,w), V(B,w))$;
- $V(\top, w) = 1$ and $V(\bot, w) = 0$;
- $V(\Box A, w) = \inf(\{1\} \cup \{V(A, w') : Rww'\})$.

A formula A is $\mathsf{GK}_\Box$-*valid*, written $\models_{\mathsf{GK}_\Box} A$, if $V(A,w) = 1$ for all Kripke models $\langle W, R, V\rangle$ for $\mathsf{GK}_\Box$ and $w \in W$.

The above definitions give a reasonable semantics for a Gödel modal logic. However, we have made a number of significant "design choices". First, note that we have omitted the dual modality $\Diamond$, characterized by the condition:

- $V(\Diamond A, w) = \sup(\{0\} \cup \{V(A, w') : Rww'\})$.

This connective is definable using negation in (classical) modal logic as $\Diamond A =_{\text{def}} \neg\Box\neg A$, but for Gödel logic, the equivalence breaks down. We concentrate here on the fuzzy logic with just one modality simply as a first step in our investigations of the full logic GK.

A second complaint may be that $\mathsf{GK}_\Box$ is not "fuzzy enough" since the accessibility relation R of a Kripke frame is Boolean-valued (crisp). Consider instead a *fuzzy Kripke frame* as a pair $\langle W, R\rangle$ where W is a non-empty set of worlds and $R : W \times W \to [0,1]$ is a binary *fuzzy accessibility relation* on W. A *fuzzy Kripke model for* $\mathsf{GK}^{\mathsf{F}}_\Box$ is then a 3-tuple $K = \langle W, R, V\rangle$ where $\langle W, R\rangle$ is a fuzzy Kripke frame and $V : Var \times W \to [0,1]$ is a mapping defined as for $\mathsf{GK}_\Box$ except that the $\Box$ condition is changed (following the semantics of Gödel implication) to:

- $V(\Box A, w) = \inf(\{1\} \cup \{V(A, w') : Rww' > V(A, w')\})$.

A formula A is $\mathsf{GK}^{\mathsf{F}}_\Box$-*valid*, written $\models_{\mathsf{GK}^{\mathsf{F}}_\Box} A$, if $V(A,w) = 1$ for all fuzzy Kripke models $\langle W, R, V\rangle$ for $\mathsf{GK}^{\mathsf{F}}_\Box$ and $w \in W$. A dual treatment of modal many-valued logics based on classical and fuzzy Kripke models is a common theme in the literature (see e.g. [11,12,5,6]). However, for this paper the distinction is not so important. It has been shown in [6] (and will follow from our results below) that in the case of $\mathsf{GK}_\Box$ and $\mathsf{GK}^{\mathsf{F}}_\Box$ the valid formulas of the two logics coincide.

Another issue is the fact that in the definition of $\Box A$, the infimum value may not be "witnessed" by the value of A at any accessible world. If we restrict to worlds where this is always the case, i.e., by insisting that every infimum is a minimum, then we get a different "witnessed" logic (see e.g. [15]). For example, the formula $\Box\neg\neg p \to \neg\neg\Box p$ (considered in [6]) is valid in the witnessed logic,

$(A1)\ A \to (B \to A)$	$(A8)\ (A \to B) \to ((C \to A) \to (C \to B))$
$(A2)\ (A \wedge B) \to A$	$(A9)\ (A \to (B \to C)) \to (B \to (A \to C))$
$(A3)\ (A \wedge B) \to B$	$(A10)\ ((A \to C) \wedge (B \to C)) \to ((A \vee B) \to C)$
$(A4)\ A \to (B \to (A \wedge B))$	$(A11)\ (A \to (B \to C)) \to ((A \wedge B) \to C)$
$(A5)\ \bot \to A$	$(A12)\ ((C \to A) \wedge (C \to B)) \to (C \to (A \wedge B))$
$(A6)\ A \to (A \vee B)$	$(A13)\ (A \to (A \to B)) \to (A \to B)$
$(A7)\ B \to (A \vee B)$	$(A14)\ (A \to B) \vee (B \to A)$

$$\frac{A \quad A \to B}{B}\ (\text{MP})$$

Fig. 1. The Hilbert System HG

but not in $\mathsf{GK}_\Box$: just consider a Kripke model $\langle \mathbb{N}, R, V \rangle$ where Rmn holds for all $m, n \in \mathbb{N}$ and $V(p, n) = 1/(n+1)$ for all $n \in \mathbb{N}$. In fact, the formula is valid in all Kripke models with a finite number of worlds, so the logic $\mathsf{GK}_\Box$ does not have the finite model property.

An axiomatization of $\mathsf{GK}_\Box$ (and $\mathsf{GK}^{\mathsf{F}}_\Box$, since the valid formulas of these logics coincide) is obtained by extending a standard axiomatization of Gödel logic such as that presented in Fig. 1 with the usual K axiom schema and also a further schema reflecting the fact, already implicit in classical logic, that $V(\neg\neg A, w) = 0$ if $V(A, w) = 0$ and 1 otherwise. More precisely, let $\mathsf{HGK}_\Box$ be HG extended with:

$$\begin{array}{ll} (K_\Box) & \Box(A \to B) \to (\Box A \to \Box B) \\ (Z_\Box) & \neg\neg\Box A \to \Box\neg\neg A \end{array} \qquad \text{and} \qquad \frac{A}{\Box A}\ (\text{NEC}).$$

Completeness for this axiomatization was established using a counter-model construction by Caicedo and Rodríguez in [6]:

Theorem 1 ([6]). $\models_{\mathsf{GK}_\Box} A$ *iff* $\models_{\mathsf{GK}^{\mathsf{F}}_\Box} A$ *iff* $\vdash_{\mathsf{HGK}_\Box} A$.

In what follows, we give an alternative proof of this theorem as a byproduct of a completeness proof for a sequent of relations calculus for $\mathsf{GK}_\Box$.

3 A Sequent of Relations Calculus

A *sequent of relations* $\mathcal{S}$ is a finite set of ordered triples:

$$A_1 \lhd_1 B_1 \mid \ldots \mid A_n \lhd_n B_n$$

where A_i and B_i are formulas and $\lhd_i \in \{<, \leq\}$ for $i = 1 \ldots n$. We say that a sequent of relations $\mathcal{S}$ is $\mathsf{GK}_\Box$-*valid*, written $\models_{\mathsf{GK}_\Box} \mathcal{S}$, if for all Kripke models $\langle W, R, V \rangle$ and $w \in W$: $V(A, w) \lhd V(B, w)$ for some $(A \lhd B) \in \mathcal{S}$.

Note that $\lhd \in \{\leq, <\}$ is used here both syntactically, as a symbol in sequents of relations, and semantically, for interpreting those sequents of relations. Let us call a sequent of relations *atomic* if it contains only propositional variables

Axioms:

$$\frac{}{\mathcal{S} \mid A \leq A}\ (\text{ID}) \qquad \frac{}{\mathcal{S} \mid A \leq \top}\ (\leq \top) \qquad \frac{}{\mathcal{S} \mid \bot \leq A}\ (\bot \leq) \qquad \frac{}{\mathcal{S} \mid \bot < \top}\ (<)$$

Structural Rules:

$$\frac{\mathcal{S} \mid A \leq B \mid C \lhd D \mid A \leq D \quad \mathcal{S} \mid A \leq B \mid C \lhd D \mid C \leq B}{\mathcal{S} \mid A \leq B \mid C \lhd D}\ (\text{COM}) \qquad \frac{\mathcal{S} \mid \top \leq \bot}{\mathcal{S}}\ (\leq) \qquad \frac{\mathcal{S}}{\mathcal{S} \mid A \lhd B}\ (\text{EW})$$

Logical Rules:

$$\frac{\mathcal{S} \mid A \lhd C \mid B \lhd C}{\mathcal{S} \mid A \wedge B \lhd C}\ (\wedge\lhd) \qquad \frac{\mathcal{S} \mid C \lhd A \quad \mathcal{S} \mid C \lhd B}{\mathcal{S} \mid C \lhd A \wedge B}\ (\lhd\wedge)$$

$$\frac{\mathcal{S} \mid A \lhd C \quad \mathcal{S} \mid B \lhd C}{\mathcal{S} \mid A \vee B \lhd C}\ (\vee\lhd) \qquad \frac{\mathcal{S} \mid C \lhd A \mid C \lhd B}{\mathcal{S} \mid C \lhd A \vee B}\ (\lhd\vee)$$

$$\frac{\mathcal{S} \mid B < A \quad \mathcal{S} \mid B < C}{\mathcal{S} \mid A \to B < C}\ (\to<) \qquad \frac{\mathcal{S} \mid A \leq B \mid C < B \quad \mathcal{S} \mid C < \top}{\mathcal{S} \mid C < A \to B}\ (<\to)$$

$$\frac{\mathcal{S} \mid \top \leq C \mid B < A \quad \mathcal{S} \mid B \leq C}{\mathcal{S} \mid A \to B \leq C}\ (\to\leq) \qquad \frac{\mathcal{S} \mid A \leq B \mid C \leq B}{\mathcal{S} \mid C \leq A \to B}\ (\leq\to)$$

Fig. 2. The Sequent of Relations Calculus SG

or constants, and *propositional* if it contains no occurrences of $\Box$. In Fig. 2, we present a sequent of relations calculus SG for G consisting of logical rules taken from [4] and some additional axioms and structural rules (based on similar calculi for G presented in [16]) for deriving valid atomic sequent of relations. As shown e.g. in [16], an atomic sequent of relations $\mathcal{S}$ is valid iff there exists $(a_i \lhd_i a_{i+1}) \in \mathcal{S}$ for $i = 1 \ldots n$ such that one of the following holds:

1. $a_1 = a_{n+1}$ or $a_1 = \bot$ or $a_{n+1} = \top$, where $\lhd_i$ is $\leq$ for some $i \in \{1, \ldots, n\}$.
2. $a_1 = \bot$ and $a_{n+1} = \top$.

It is an easy induction to show that an atomic sequent of relations $\mathcal{S}$ is valid iff it is derivable using the axioms and structural rules of SG (see e.g. [16] for very similar proofs). Since also the logical rules of SG are sound and invertible (see e.g. [16]), it follows that:

Theorem 2. $\models_{\mathsf{GK}_\Box} \mathcal{S}$ *iff* $\vdash_{\mathsf{SG}} \mathcal{S}$ *for any propositional sequent of relations* $\mathcal{S}$.

The sequent of relations calculus $\mathsf{SGK}_\Box$ consists of SG extended with the rules:

$$\frac{A_1 \leq B \mid \ldots \mid A_n \leq B \mid C_1 \leq \bot \mid \ldots \mid C_m \leq \bot}{\mathcal{S} \mid \Box A_1 \leq \Box B \mid \ldots \mid \Box A_n \leq \Box B \mid \Box C_1 \leq \bot \mid \ldots \mid \Box C_m \leq \bot}\ (\Box) \quad (n \geq 1, m \geq 0)$$

$$\frac{\mathcal{S} \mid A \leq B \mid \Box\top \leq B}{\mathcal{S} \mid A \leq B}\ (\text{WL}) \qquad \frac{\mathcal{S} \mid A \leq B \mid A \leq \bot}{\mathcal{S} \mid A \leq B}\ (\text{WR})$$

Note that the rather ugly "weakening" rules (WL) and (WR) allow us to work backwards from relations of the form $A \leq \Box B$ and $\Box A \leq B$ to $\Box\top \leq \Box B$ and $\Box A \leq \bot$, respectively, so that they fit the format of the rule $(\Box)$.

Example 1. Consider the following derivation of a sequent of relations corresponding to the K axiom schema $\Box(A \to B) \to (\Box A \to \Box B)$:

$$
\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{A \leq B \mid B < A \mid A \leq A}\,(\textsc{id}) \quad \dfrac{}{A \leq B \mid B < A \mid B \leq B}\,(\textsc{id})}{A \leq B \mid B < A}\,(\textsc{com})}{A \leq B \mid \top \leq B \mid B < A}\,(\textsc{ew}) \qquad \dfrac{}{A \leq B \mid B \leq B}\,(\textsc{id})}{A \leq B \mid A \to B \leq B}\,(\to\!\leq)}{\Box A \leq \Box B \mid \Box(A \to B) \leq \Box B}\,(\Box)}{\Box(A \to B) \leq \Box A \to \Box B}\,(\leq\!\to)}{\Box(A \to B) \leq \Box A \to \Box B \mid \top \leq \Box A \to \Box B}\,(\textsc{ew})}{\top \leq \Box(A \to B) \to (\Box A \to \Box B)}\,(\leq\!\to)
$$

Theorem 3. *If* $\vdash_{\mathsf{SGK}_\Box} \mathcal{S}$*, then* $\models_{\mathsf{GK}_\Box} \mathcal{S}$*.*

Proof. The proof is a standard induction on the height of a derivation of $\mathcal{S}$ in $\mathsf{SGK}_\Box$; the only significantly new case is to establish the soundness of $(\Box)$. Suppose that there is a counter-model $K = \langle W, R, V\rangle$ for the conclusion. I.e. for some $w \in W$: $V(\Box A_i, w) > V(\Box B, w)$ for $i = 1 \ldots n$ and $V(\Box C_j, w) > 0$ for $j = 1 \ldots m$. Then there must be a world w' accessible to w where $V(A_i, w') > V(B, w')$ for $i = 1 \ldots n$. Moreover, since $V(\Box C_j, w) \neq 0$, we must also have $V(C_j, w') > 0$ for $j = 1 \ldots m$ as required. $\Box$

Completeness is of course more complicated. Notice first that all the logical rules (as proved in [4]), (COM), (WL), and (WR) are invertible in the sense that if the conclusion is $\mathsf{GK}_\Box$-valid, then so are the premises. Hence applying the logical rules backwards to a $\mathsf{GK}_\Box$-valid sequent of relations results in $\mathsf{GK}_\Box$-valid sequent of relations containing only modal formulas and atoms. Let us call a sequent of relations $\mathcal{S}$ containing only modal formulas and atoms *saturated* if whenever $\mathcal{S}$ occurs as the conclusion of (COM), $(\leq)$, (WL), or (WR), then $\mathcal{S}$ also occurs as one of the premises. In other words, the sequent of relations is "closed" under applications of these rules. Moreover, since sequents of relations are sets of pairs of formulas, there is a finite number that can be obtained by applying the rules backwards to any given sequent of relations. Hence easily:

Lemma 1. *Every sequent of relations* $\mathcal{S}$ *is derivable from a set of saturated sequent of relations* $\mathcal{S}_1, \ldots, \mathcal{S}_n$ *using the logical rules,* (COM)*,* $(\leq)$*,* (WL)*, and* (WR)*, and if* $\models_{\mathsf{GK}_\Box} \mathcal{S}$*, then* $\models_{\mathsf{GK}_\Box} \mathcal{S}_i$ *for* $i = 1 \ldots n$*.*

The challenge now is to show that a valid saturated sequent of relations $\mathcal{S}$ is derivable. Our strategy will be to use Lemmas 2 and 4 to show that *either* $\mathcal{S}$ is derivable using just the rules of SG *or* the part of $\mathcal{S}$ containing only relations of the form $\Box A \leq \Box B$ and $\Box C \leq \bot$ is itself valid. For the latter case, $\mathcal{S}$ is derivable using $(\Box)$ from a less complex sequent of relations that is proved valid in Lemma 5. An inductive argument then completes the proof.

Let us say that a formula *occurs* in a sequent of relations if it occurs as the left or right side of one of the relations. Then a sequent of relations is said to be *propositionally* $\mathsf{GK}_\Box$*-valid* if the sequent of relations obtained by replacing each occurrence of a modal formula $\Box A$ with a variable p_A is $\mathsf{GK}_\Box$-valid.

Lemma 2. *Let $\mathcal{S} \mid \mathcal{S}'$ be saturated where $\mathcal{S}$ contains only relations of the form $\Box A \triangleleft \Box B$, $\Box A \leq \bot$, $\Box A < \top$, and $\bot < \Box B$, and $\mathcal{S}'$ contains no relations of this form. If $\models_{\mathsf{GK}_\Box} \mathcal{S} \mid \mathcal{S}'$, then either $\mathcal{S} \mid \mathcal{S}'$ is propositionally $\mathsf{GK}_\Box$-valid or $\models_{\mathsf{GK}_\Box} \mathcal{S}$.*

Proof. Proceeding contrapositively, suppose that $\not\models_{\mathsf{GK}_\Box} \mathcal{S}$ and $\mathcal{S} \mid \mathcal{S}'$ is *not* propositionally $\mathsf{GK}_\Box$-valid. Then for some model $K = \langle W, R, V\rangle$ and $w \in W$:

(1) $V(\Box A, w) \not\triangleleft V(\Box B, w)$ for all $(\Box A \triangleleft \Box B) \in \mathcal{S}$.
(2) $V(\Box A, w) > 0$ for all $(\Box A \leq \bot) \in \mathcal{S}$.
(3) $V(\Box A, w) = 1$ for all $(\Box A < \top) \in \mathcal{S}$.
(4) $V(\Box B, w) = 0$ for all $(\bot < \Box B) \in \mathcal{S}$.

For each $\Box A$ occurring in $\mathcal{S} \mid \mathcal{S}'$, let us add a constant c_A to the language so that for any model $\langle W', R', V'\rangle$ and world $x \in W'$:

$$V'(c_A, x) = V(\Box A, w).$$

Let $(\mathcal{S} \mid \mathcal{S}')^P$ be $\mathcal{S} \mid \mathcal{S}'$ with each $\Box A$ occurring in $\mathcal{S} \mid \mathcal{S}'$ replaced by c_A.

Claim: $\not\models_{\mathsf{GK}_\Box} (\mathcal{S} \mid \mathcal{S}')^P$.

The result follows from this claim. Let $v : \mathit{Var} \to [0, 1]$ be the propositional counter-valuation for $(\mathcal{S} \mid \mathcal{S}')^P$. Define $K' = \langle W \cup \{w_0\}, R', V'\rangle$ where:

1. $R' = R \cup \{(w_0, w') : (w, w') \in R\}$.
2. V' is V extended with $V'(p, w_0) = v(p)$ for all variables p.

Then $V'(\Box A, w_0) = V(\Box A, w)$ for all $\Box A$ occurring in $\mathcal{S} \mid \mathcal{S}'$. So, since v is a counter-valuation for $(\mathcal{S} \mid \mathcal{S}')^P$, we have $\not\models_{\mathsf{GK}_\Box} \mathcal{S} \mid \mathcal{S}'$ as required.

Proof of claim. Proceeding by contraposition, suppose that $\models_{\mathsf{GK}_\Box} (\mathcal{S} \mid \mathcal{S}')^P$. Then using (1)-(4), $\models_{\mathsf{GK}_\Box} (\mathcal{S}')^P$. By saturation, there are several possibilities:

(i) $(\mathcal{S}')^P$ contains $a \leq a$ or $\bot \leq a$ or $a \leq \top$ or $\bot < \top$. But then $\mathcal{S}'$ is propositionally $\mathsf{GK}_\Box$-valid, a contradiction.
(ii) $(\mathcal{S}')^P$ contains $c_C \triangleleft c_D$ or $c_C < \top$ or $\bot < c_D$. But then $\mathcal{S}'$ contains $\Box C \triangleleft \Box D$ or $\Box C < \top$ or $\bot < \Box D$, a contradiction.
(iii) $(\mathcal{S}')^P$ contains $a \leq c_D$ and $V(\Box D, w) = 1$. But then by (WL), $\mathcal{S}$ contains $\Box\top \leq \Box D$ and $V(\Box D, w) < 1$, a contradiction.
(iv) $(\mathcal{S}')^P$ contains $c_C \leq a$ and $V(\Box C, w) = 0$. But then by (WR), $\mathcal{S}$ contains $\Box C \leq \bot$ and $V(\Box C, w) > 0$, a contradiction. □

The next step is to remove relations involving strict inequalities. We make use of the following scaling lemmas, easily proved by induction on formula complexity.

Lemma 3. *Let $K = \langle W, R, V\rangle$ be a model and $c \in (0, 1)$.*

(a) Let $K' = \langle W, R, V'\rangle$ where for each variable p and $x \in W$:

$$V'(p, x) = \begin{cases} 1 & \text{if } V(p, x) \geq c \\ V(p, x) & \text{otherwise.} \end{cases}$$

Then for all $x \in W$ and every formula A:

$$V'(A,x) = \begin{cases} 1 & \text{if } V(A,x) \geq c \\ V(A,x) & \text{otherwise.} \end{cases}$$

(b) Let $K' = \langle W, R, V' \rangle$ where $0 < a < b < 1$ and for each variable p and $x \in W$:

$$V'(p,x) = \begin{cases} a + c(V(p,x) - a) & \text{if } V(p,x) \in (a,b] \\ V(p,x) & \text{otherwise.} \end{cases}$$

Then for all $x \in W$ and every formula A:

$$V'(A,x) = \begin{cases} a + c(V(A,x) - a) & \text{if } V(A,x) \in (a,b] \\ V(p,x) & \text{otherwise.} \end{cases}$$

Lemma 4. *Let $\mathcal{S} \mid F < G$ be saturated and contain only relations of the form $\Box A \lhd \Box B$, $\Box A \leq \bot$, $\Box A < \top$, and $\bot < \Box B$. If $\models_{\mathsf{GK}_\Box} \mathcal{S} \mid F < G$, then $\models_{\mathsf{GK}_\Box} \mathcal{S}$.*

Proof. Let us consider just the case where $F < G$ is of the form $\Box A < \Box B$, since the other cases are very similar. Proceeding contrapositively, suppose that $\not\models_{\mathsf{GK}_\Box} \mathcal{S}$. Hence there is a model $K = \langle W, R, V \rangle$ and $w \in W$ such that:

(1) $V(\Box C, w) \not\lhd V(\Box D, w)$ for all $(\Box C \lhd \Box D) \in \mathcal{S}$.
(2) $V(\Box C, w) > 0$ for all $(\Box C \leq \bot) \in \mathcal{S}$.
(3) $V(\Box C, w) = 1$ for all $(\Box C < \top) \in \mathcal{S}$.
(4) $V(\Box D, w) = 0$ for all $(\bot < \Box D) \in \mathcal{S}$.

If $V(\Box A, w) \geq V(\Box B, w)$, then $\not\models_{\mathsf{GK}_\Box} \mathcal{S} \mid \Box A < \Box B$ as required, so assume that:

(5) $V(\Box A, w) < V(\Box B, w)$.

Since $\mathcal{S} \mid F < G$ is saturated, for each $(\Box C \leq \Box D) \in \mathcal{S}$:

either $(\Box C \leq \Box B) \in \mathcal{S}$ and, by (1), $V(\Box C, w) > V(\Box B, w)$
or $(\Box A \leq \Box D) \in \mathcal{S}$ and, by (1), $V(\Box A, w) > V(\Box D, w)$

and in particular:

(6) $V(\Box A, w) \leq V(\Box D, w) < V(\Box C, w) \leq V(\Box B, w)$ is not possible.

We have two cases:

(i) Suppose that $V(\Box B, w) < 1$. Then using Lemma 3 (b), we define for each $i \in \mathbb{Z}^+$, a counter-model $K_i = \langle W_i, R_i, V_i \rangle$ where:
   1. $\langle W_i, R_i \rangle$ is a copy of $\langle W, R \rangle$ with distinct worlds for each $i \in \mathbb{Z}^+$ and w_i is the corresponding copy of w.
   2. For all formulas E satisfying $V_i(\Box A, w_i) < V_i(E, w_i) \leq V_i(\Box B, w_i)$: $V_i(\Box A, w_i) < V_i(E, w_i) < V(\Box A, w_i) + 1/i$.

   Now we define a model $K' = \langle W', R', V' \rangle$ where:

1. $W' = \{w_0\} \cup \bigcup_{i \in \mathbb{Z}^+} W_i$.
2. $R' = \{(w_0, w') : (w_i, w') \in V_i \text{ for some } i \in \mathbb{Z}^+\} \cup \bigcup_{i \in \mathbb{Z}^+} R_i$.
3. $V'(p, x) = V_i(p, x)$ for all $x \in W_i$ and $V'(p, w_0) = 0$.

But then:
$$\begin{aligned} V'(\Box B, w_0) &= \inf(\{1\} \cup \{V'(B, w') : R'w_0w'\}) \\ &= \inf\{V_i(\Box B, w_i) : i \in \mathbb{Z}^+\} \\ &= V'(\Box A, w_0). \end{aligned}$$

If $V(C, w) \geq V(D, w)$ for some $\Box C < \Box D \in \mathcal{S}$, then $V'(\Box C, w_0) \geq V'(\Box D, w_0)$. If $\Box C \leq \Box D \in \mathcal{S}$, then by (6), it follows that $V'(\Box C, w_0) > V'(\Box D, w_0)$.

(ii) Now suppose that $V(\Box B, w) = 1$. Then using Lemma 3 (a), we obtain a model $K' = \langle W, R, V' \rangle$ where $V'(\Box A, w) = V'(\Box B, w) = 1$. As in case (i), again by (6), the other inequalities required are not affected. □

Lemma 5. *Suppose that* $\models_{\mathsf{GK}_\Box} \{\Box A_i \leq \Box B_i\}_{i=1}^n \mid \{\Box C_j \leq \bot\}_{j=1}^m$. *Then* $\models_{\mathsf{GK}_\Box} \bigwedge_{i \in I} A_i \leq \bigwedge_{i \in I} B_i \mid \{C_j \leq \bot\}_{j=1}^m$ *for some* $\emptyset \subset I \subseteq \{1, \ldots, n\}$.

Proof. We argue by contraposition; i.e., suppose that:

$$\not\models_{\mathsf{GK}_\Box} \bigwedge_{i \in I} A_i \leq \bigwedge_{i \in I} B_i \mid \{C_j \leq \bot\}_{j=1}^m \quad \text{for all} \quad \emptyset \subset I \subseteq \{1, \ldots, n\}.$$

We obtain a model for each $i \in \{1, \ldots, n\}$ as follows. By assumption:

$$\not\models_{\mathsf{GK}_\Box} A_i \wedge \ldots \wedge A_n \leq B_i \wedge \ldots \wedge B_n \mid \{C_j \leq \bot\}_{j=1}^m.$$

So we have $K_i = \langle W_i, R_i, V_i \rangle$ and $x_i \in W_i$ (with each W_i distinct) such that:

$$V_i(A_i \wedge \ldots \wedge A_n, x_i) > V_i(B_i \wedge \ldots \wedge B_n, x_i) \quad \text{and} \quad V_i(C_j) > 0 \quad \text{for } j = 1 \ldots m.$$

Moreover, without loss of generality we can assume:

$$V_i(B_i, x_i) \leq V_i(B_k, x_i) \quad \text{and so} \quad V_i(A_k, x_i) > V_i(B_i, x_i) \quad \text{for } k = i \ldots n.$$

Now using Lemma 3, we define iteratively $K'_i = \langle W_i, R_i, V'_i \rangle$ for $i = n \ldots 1$ such that for $j = i \ldots n$:

(1) $V'_j(B_j, x_j) < V_k(A_j, x_k)$ for $k = 1 \ldots i - 1$.
(2) $V'_j(B_j, x_j) < V'_k(A_j, x_k)$ for $k = i \ldots n$.
(3) $V'_j(C_k, x_j) > 0$ for $k = 1 \ldots m$.

We achieve this by scaling the interval $[0, V_i(B_i, x_i)]$ (if necessary) to the smaller interval $[0, V'_i(B_i, x_i)]$ so that (1) and (2) hold with j replaced by i. Now consider $j \in \{i, \ldots, n\}$. Clearly $V'_i(B_j, x_j) \leq V_i(B_j, x_j)$. Moreover, since $V_i(A_j, x_i) > V_i(B_i, x_i)$, also $V'_i(A_j, x_i) = V_i(A_j, x_i) > V'_j(B_j, x_j)$ as required.

Finally, we define a model $K = \langle W, R, V \rangle$ where for a new world w_0:

1. $W = W_1 \cup \ldots \cup W_n \cup \{w_0\}$.
2. $R = R_1 \cup \ldots \cup R_n \cup \{(w_0, w) : (x_i, w) \in R_i \text{ for some } i \in \{1, \ldots, n\}\}$.
3. $V(p, x) = V'_i(p, x)$ for all $x \in W_i$ and $V(p, w_0) = 0$.

Then by (1)-(3) above, $V(\Box B_i, x_0) = V(B_i, x_i) < V(\Box A_i, x_0)$ for $i = 1 \ldots n$ and $V(\Box C_k, x_0) > 0$ for $k = 1 \ldots m$ as required. $\Box$

Theorem 4 (Completeness). *If* $\models_{\mathsf{GK}_\Box} \mathcal{S}$*, then* $\vdash_{\mathsf{SGK}_\Box} \mathcal{S}$*.*

Proof. We prove the theorem by induction on the modal degree of the sequent of relations $\mathcal{S}$: the maximal complexity of a boxed subformula occurring in $\mathcal{S}$. If the modal degree is 0, then $\mathcal{S}$ is propositional and $\mathsf{SGK}_\Box$-derivable. Otherwise, by Lemma 1 (since working upwards, the rules do not increase modal degree), we can assume that $\mathcal{S}$ is both $\mathsf{GK}_\Box$-valid and saturated. If $\mathcal{S}$ is propositionally $\mathsf{GK}_\Box$-valid, then it is derivable. Otherwise, by Lemmas 2 and 4, $\mathcal{S}$ is of the form:

$$\mathcal{S}' \mid \{\Box A_i \leq \Box B_i\}_{i=1}^n \mid \{\Box C_j \leq \bot\}_{j=1}^m$$

and $\models_{\mathsf{GK}_\Box} \{\Box A_i \leq \Box B_i\}_{i=1}^n \mid \{\Box C_j \leq \bot\}_{j=1}^m$. But then by Lemma 5:

$$\models_{\mathsf{GK}_\Box} \bigwedge_{i \in I} A_i \leq \bigwedge_{i \in I} B_i \mid \{C_j \leq \bot\}_{j=1}^m \quad \text{for some } \emptyset \subset I \subseteq \{1, \ldots, n\}.$$

Let us assume without loss of generality that $I = \{1, \ldots, n\}$. Then also:

$$\models_{\mathsf{GK}_\Box} \{A_i \leq B_k\}_{i=1}^n \mid \{C_j \leq \bot\}_{j=1}^m \quad \text{for } k = 1 \ldots n.$$

So by the induction hypothesis and an application of ($\Box$):

$$\vdash_{\mathsf{SGK}} \mathcal{S}' \mid \{\Box A_i \leq \Box B_k\}_{i=1}^n \mid \{\Box C_j \leq \bot\}_{j=1}^m \quad \text{for } k = 1 \ldots n.$$

But then $\mathcal{S}$ is derivable by repeated applications of (COM). $\Box$

We can use $\mathsf{SGK}_\Box$ to give an alternative completeness proof for the axiomatization $\mathsf{HGK}_\Box$ with respect to both standard and fuzzy Kripke frames (Theorem 1).

Corollary 1. $\models_{\mathsf{GK}_\Box} A$ *iff* $\models_{\mathsf{GK}^{\mathsf{F}}_\Box} A$ *iff* $\vdash_{\mathsf{HGK}_\Box} A$*.*

Proof. First notice that the soundness proof for $\mathsf{SGK}_\Box$ can be adjusted to prove soundness for fuzzy frames: i.e., if $\vdash_{\mathsf{SGK}_\Box} \mathcal{S}$, then $\models_{\mathsf{GK}^{\mathsf{F}}_\Box} \mathcal{S}$. But then it follows from the completeness proof (the other direction is trivial) that $\models_{\mathsf{GK}^{\mathsf{F}}_\Box} \mathcal{S}$ iff $\models_{\mathsf{GK}_\Box} \mathcal{S}$. Now consider the following interpretation of sequent of relations:

$$I(\{A_i < B_i\}_{i=1}^n \mid \{C_j \leq D_j\}_{j=1}^m) = \bigwedge_{i=1}^n (B_i \to A_i) \to \bigvee_{j=1}^m (C_j \to D_j).$$

It is easily shown that $\models_{\mathsf{GK}_\Box} \mathcal{S}$ iff $\models_{\mathsf{GK}_\Box} I(\mathcal{S})$. Hence to establish completeness for $\mathsf{HGK}_\Box$, it is sufficient to show that for each rule $\mathcal{S}_1, \ldots, \mathcal{S}_n \ / \ \mathcal{S}$ of $\mathsf{SGK}_\Box$, whenever $\vdash_{\mathsf{HGK}_\Box} I(\mathcal{S}_i)$ for $i = 1 \ldots n$, also $\vdash_{\mathsf{HGK}_\Box} I(\mathcal{S})$. This is straightforward for the propositional rules and easy for the weakening rules, so let us just consider $\Box$. We show the simplified case that if $\vdash_{\mathsf{HGK}_\Box} (A \to B) \vee \neg C$, then $\vdash_{\mathsf{HGK}_\Box} (\Box A \to \Box B) \vee \neg \Box C$. Note first that $\vdash_{\mathsf{HG}} (\neg\neg F \to G) \leftrightarrow (G \vee \neg F)$. Hence if $\vdash_{\mathsf{HGK}_\Box} (A \to B) \vee \neg C$, then $\vdash_{\mathsf{HGK}_\Box} \neg\neg C \to (A \to B)$. But then by (NEC), $\vdash_{\mathsf{HGK}_\Box} \Box(\neg\neg C \to (A \to B))$ and using ($K_\Box$), $\vdash_{\mathsf{HGK}_\Box} \Box\neg\neg C \to (\Box A \to \Box B)$. Now, using ($Z_\Box$), $\vdash_{\mathsf{HGK}_\Box} \neg\neg\Box C \to (\Box A \to \Box B)$. But then $\vdash_{\mathsf{HGK}_\Box} (\Box A \to \Box B) \vee \neg \Box C$ as required. $\Box$

Theorem 5. *The* $\mathsf{GK}_\Box$*-validity problem for* $\mathsf{GK}_\Box$ *is PSPACE-complete.*

Proof. First we show that $\mathsf{GK}_\Box$ is PSPACE-hard. We recall that the modal logic K is PSPACE-complete (see e.g. [7]). Consider the translation $*$ sending each propositional variable p to its double negation $\neg\neg p$. We can easily show that $\models_{\mathsf{K}} A$ iff $\models_{\mathsf{GK}_\Box} A^*$ which establishes that $\mathsf{GK}_\Box$ must also be PSPACE-hard. For the non-trivial direction consider any GK model $M = \langle W, R, V\rangle$ and define a K model $M' = \langle W, R, V'\rangle$ by stipulating: $V'(p, w) = V(\neg\neg p, w)$. Then by a simple induction, $V'(C, w) = V(C^*, w) \in \{0, 1\}$ for any formula C. Thus if $\not\models_{\mathsf{GK}_\Box} A^*$, then there is a K model M where A is not valid.

For PSPACE-inclusion, we consider the sequent of relations calculus $\mathsf{SGK}_\Box$. Given a formula A, let $Sub(A)$ be the set of subformulas of A together with the formulas $\top$, $\Box\top$, $\bot$, $\Box\bot$, and consider the set

$$\Phi_A = \{C \lhd D : C, D \in Sub(A), \lhd \in \{<, \leq\}\}.$$

The cardinality of Φ_A is $O(|A|^2)$. Since any sequent of relations appearing in a derivation of $\top \leq A$ is a subset of Φ_A, its size is also $O(|A|^2)$.

We now consider the length of branches in the search for an $\mathsf{SGK}_\Box$-derivation of $\top \leq A$. Using the invertibility of the logical rules we assume that any branch of a derivation is expanded by applying iteratively the rules upwards in the following order:

(1) Apply the logical rules, (WL), (WR), ($\leq$), and (COM) in order to obtain a saturated sequent and check the axioms.
(2) Apply ($\Box$) and restart from (1) with the premise of this rule.

The length of the branch built in (1) is $O(|A|^2)$ since each logical rule replaces one relation with one or two relations involving formulas of smaller complexity, and each application of (WL), (WR), ($\leq$), and (COM) add exactly one relation at a time, with the total number of different relations possible being $O(|A|^2)$. The sequent obtained in (2) by applying ($\Box$) has a smaller or equal number of relations and a strictly smaller modal degree. The entire length of a proof branch is hence bounded by $O(|A|^2 \times m) = O(|A|^3)$, where m is the modal degree of A.

Thus storing a branch of a proof requires only polynomial space. Moreover, the branching is at most binary. As usual, we search for a proof in a depth-first manner: we store one branch at a time together with some information (requiring a small amount of space, say logarithmic space) to reconstruct branching points and backtracking points, the latter determined by alternative applications of ($\Box$). Hence the total amount of space needed for carrying out proof search is polynomial in the size of A, and so deciding validity for $\mathsf{GK}_\Box$ is in PSPACE. $\Box$

4 A Hypersequent Calculus

Hypersequents were introduced by Avron in [1] as a generalization of Gentzen sequents that allow disjunctive or parallel forms of reasoning. Instead of a single sequent, there is a collection of sequents that can be "worked on" simultaneously.

Initial Hypersequents

$$\frac{}{\mathcal{G} \mid A \Rightarrow A}\ (\text{ID}) \qquad \frac{}{\mathcal{G} \mid \Gamma, \bot \Rightarrow \Delta}\ (\bot\Rightarrow) \qquad \frac{}{\mathcal{G} \mid \Gamma \Rightarrow \top}\ (\Rightarrow\top)$$

Structural Rules

$$\frac{\mathcal{G}}{\mathcal{G} \mid \mathcal{H}}\ (\text{EW}) \qquad \frac{\mathcal{G} \mid \mathcal{H} \mid \mathcal{H}}{\mathcal{G} \mid \mathcal{H}}\ (\text{EC}) \qquad \frac{\mathcal{G} \mid \Gamma_1, \Pi_1 \Rightarrow \Delta_1 \quad \mathcal{G} \mid \Gamma_2, \Pi_2 \Rightarrow \Delta_2}{\mathcal{G} \mid \Gamma_1, \Gamma_2 \Rightarrow \Delta_1 \mid \Pi_1, \Pi_2 \Rightarrow \Delta_2}\ (\text{COM})$$

$$\frac{\mathcal{G} \mid \Gamma \Rightarrow \Delta}{\mathcal{G} \mid \Gamma, A \Rightarrow \Delta}\ (\text{WL}) \qquad \frac{\mathcal{G} \mid \Gamma \Rightarrow}{\mathcal{G} \mid \Gamma \Rightarrow A}\ (\text{WR}) \qquad \frac{\mathcal{G} \mid \Gamma, A, A \Rightarrow \Delta}{\mathcal{G} \mid \Gamma, A \Rightarrow \Delta}\ (\text{CL})$$

Logical Rules

$$\frac{\mathcal{G} \mid \Gamma_1 \Rightarrow A \quad \mathcal{G} \mid \Gamma_2, B \Rightarrow \Delta}{\mathcal{G} \mid \Gamma_1, \Gamma_2, A \to B \Rightarrow \Delta}\ (\to\Rightarrow) \qquad \frac{\mathcal{G} \mid \Gamma, A \Rightarrow B}{\mathcal{G} \mid \Gamma \Rightarrow A \to B}\ (\Rightarrow\to)$$

$$\frac{\mathcal{G} \mid \Gamma, A \Rightarrow \Delta}{\mathcal{G} \mid \Gamma, A \wedge B \Rightarrow \Delta}\ (\wedge\Rightarrow)_1 \qquad \frac{\mathcal{G} \mid \Gamma, B \Rightarrow \Delta}{\mathcal{G} \mid \Gamma, A \wedge B \Rightarrow \Delta}\ (\wedge\Rightarrow)_2 \qquad \frac{\mathcal{G} \mid \Gamma \Rightarrow A \quad \mathcal{G} \mid \Gamma \Rightarrow B}{\mathcal{G} \mid \Gamma \Rightarrow A \wedge B}\ (\Rightarrow\wedge)$$

$$\frac{\mathcal{G} \mid \Gamma, A \Rightarrow \Delta \quad \mathcal{G} \mid \Gamma, B \Rightarrow \Delta}{\mathcal{G} \mid \Gamma, A \vee B \Rightarrow \Delta}\ (\vee\Rightarrow) \qquad \frac{\mathcal{G} \mid \Gamma \Rightarrow A}{\mathcal{G} \mid \Gamma \Rightarrow A \vee B}\ (\Rightarrow\vee)_1 \qquad \frac{\mathcal{G} \mid \Gamma \Rightarrow B}{\mathcal{G} \mid \Gamma \Rightarrow A \vee B}\ (\Rightarrow\vee)_2$$

Cut Rule

$$\frac{\mathcal{G} \mid \Gamma_1, A \Rightarrow \Delta \quad \mathcal{G} \mid \Gamma_2 \Rightarrow A}{\mathcal{G} \mid \Gamma_1, \Gamma_2 \Rightarrow \Delta}\ (\text{CUT})$$

Fig. 3. The Gentzen System GG

More precisely, we define a *sequent* S here as an ordered pair consisting of a finite multiset Γ of formulas and a multiset Δ containing at most one formula, written $\Gamma \Rightarrow \Delta$, and a *hypersequent* $\mathcal{G}$ as a non-empty finite multiset of sequents, written $\Gamma_1 \Rightarrow \Delta_1 \mid \ldots \mid \Gamma_n \Rightarrow \Delta_n$ or sometimes, for short, as $[\Gamma_i \Rightarrow \Delta_i]_{i=1}^n$. By taking multisets of formulas and sequents rather than sequences (as used e.g. by Avron in [1]) or sets we ensure that the multiplicity but not the order of elements is important. We interpret sequents and hypersequents as follows (recalling that $\bigwedge[] =_{\text{def}} \top$ and $\bigvee[] =_{\text{def}} \bot$):

$$\mathrm{I}(\Gamma \Rightarrow \Delta) =_{\text{def}} \bigwedge \Gamma \to \bigvee \Delta \qquad \mathrm{I}(S_1 \mid \ldots \mid S_n) =_{\text{def}} \mathrm{I}(S_1) \vee \ldots \vee \mathrm{I}(S_n).$$

Hypersequent calculi admitting cut-elimination have been defined for a wide range of fuzzy logics (for details see [16]). In particular, the first example of such a system was the calculus GG defined for Gödel logic by Avron in [2] (see also [3,16]). We extend this system, presented in Figure 3 with an adapted version of the usual Gentzen rule for K. $\mathsf{GGK}_\Box$ is GG extended with:

$$\frac{\Pi \Rightarrow \mid \Gamma \Rightarrow A}{\Box\Pi \Rightarrow \mid \Box\Gamma \Rightarrow \Box A}\ (\Box)$$

Rules for the defined negation $\neg A =_{\text{def}} A \to \bot$ are derivable in $\mathsf{GGK}_\Box$:

$$\frac{\mathcal{G} \mid \Gamma \Rightarrow A}{\mathcal{G} \mid \Gamma, \neg A \Rightarrow}\ (\neg\Rightarrow) \qquad \frac{\mathcal{G} \mid \Gamma, A \Rightarrow}{\mathcal{G} \mid \Gamma \Rightarrow \neg A}\ (\Rightarrow\neg)$$

It will also be helpful (e.g. in proving cut-elimination) to consider a generalization of the rule ($\Box$), derivable using (COM), ($\Box$), (CL), and (WL):

$$\frac{\Pi_1 \Rightarrow | \ldots | \Pi_n \Rightarrow | \Gamma \Rightarrow A}{\Box\Pi_1 \Rightarrow | \ldots | \Box\Pi_n \Rightarrow | \Box\Gamma \Rightarrow \Box A}\ (\Box)^* \qquad (n \in \mathbb{N})$$

Example 2. All the axioms of $\mathsf{HGK}_\Box$ are derivable in $\mathsf{GGK}_\Box$; e.g. for ($Z_\Box$):

$$\begin{array}{c}
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We proceed by induction on the lexicographically ordered pair $\langle |A|, |d_1| + |d_2| \rangle$. If the last step in either derivation is an initial hypersequent, then the result follows almost immediately. Also, if the last step in either derivation is not ($\Box$) or does not have the cut-formula as the principal formula, then the result follows by applications of the induction hypothesis to the premises and applications of the same rule and structural rules. Suppose for example that one of the derivations ends with an application of (COM) (the other derivation may end with ($\Box$)):

$$\frac{\mathcal{G} \mid \Gamma_1', \Gamma_2', [A]^{\lambda_1'+\lambda_2'} \Rightarrow \Delta_1 \quad \mathcal{G} \mid \Gamma_1'', \Gamma_2'', [A]^{\lambda_1''+\lambda_2''} \Rightarrow \Delta_2}{\mathcal{G} \mid \Gamma_1', \Gamma_1'', [A]^{\lambda_1'+\lambda_1''} \Rightarrow \Delta_1 \mid \Gamma_2', \Gamma_2'', [A]^{\lambda_2'+\lambda_2''} \Rightarrow \Delta_2}$$

where $\mathcal{G} = [\Gamma_i, [A]^{\lambda_i} \Rightarrow \Delta_i]_{i=3}^{n}$. Then by the induction hypothesis twice:

$$\vdash_{\mathsf{GGK}^{\circ}_{\Box}} \mathcal{H}' \mid [\Gamma_1', \Gamma_2', \Pi_j^{\lambda_1'+\lambda_2'} \Rightarrow \Delta_1]_{j=1}^{m} \quad \vdash_{\mathsf{GGK}^{\circ}_{\Box}} \mathcal{H}' \mid [\Gamma_1'', \Gamma_2'', \Pi_j^{\lambda_1''+\lambda_2''} \Rightarrow \Delta_2]_{j=1}^{m}$$

where $\mathcal{H}' = \mathcal{H} \mid [\Gamma_i, \Pi_j^{\lambda_i} \Rightarrow \Delta_i]_{i=3\ldots n}^{j=1\ldots m}$. The required hypersequent:

$$\mathcal{H}' \mid [\Gamma_1', \Pi_j^{\lambda_1'+\lambda_1''}, \Gamma_1'' \Rightarrow \Delta_1]_{j=1}^{m} \mid [\Gamma_2', \Pi_j^{\lambda_2'+\lambda_2''}, \Gamma_2'' \Rightarrow \Delta_2]_{j=1}^{m}$$

is derivable by repeated applications of (COM), (EC), and (EW).

If a distinguished occurrence of A is the principal formula in both derivations and of the form $B \wedge C$, $B \vee C$, or $B \rightarrow C$, then we can first use the induction hypothesis applied to the premises in one derivation and the conclusion in the other, and then apply the induction hypothesis again with cut-formulas B and C of smaller complexity. The result follows using applications of (EC) and/or (EW) as required. Consider then the hardest case where both derivations d_1 and d_2 end as follows with an application of ($\Box$) and A is of the form $\Box B$:

$$\frac{\dfrac{\vdots}{\Gamma_1, [B]^{\lambda_1} \Rightarrow \mid \Gamma_2, [B]^{\lambda_2} \Rightarrow C}}{\Box\Gamma_1, [\Box B]^{\lambda_1} \Rightarrow \mid \Box\Gamma_2, [\Box B]^{\lambda_2} \Rightarrow \Box C}\ (\Box) \qquad \frac{\dfrac{\vdots}{\Sigma \Rightarrow \mid \Pi \Rightarrow B}}{\Box\Sigma \Rightarrow \mid \Box\Pi \Rightarrow \Box B}\ (\Box)$$

Then since $|B| < |\Box B|$, we can apply the induction hypothesis to the $\mathsf{GGK}^{\circ}_{\Box}$-derivable hypersequents $\Gamma_1, [B]^{\lambda_1} \Rightarrow \mid \Gamma_2, [B]^{\lambda_2} \Rightarrow C$ and $\Sigma \Rightarrow \mid \Pi \Rightarrow B$ to obtain a $\mathsf{GGK}^{\circ}_{\Box}$-derivation of $\Sigma \Rightarrow \mid \Gamma_1, \Pi^{\lambda_1} \Rightarrow \mid \Gamma_2, \Pi^{\lambda_2} \Rightarrow C$. Hence by an application of the derived rule $(\Box)^*$, we obtain a $\mathsf{GGK}^{\circ}_{\Box}$-derivation ending with $\Box\Sigma \Rightarrow \mid \Box\Gamma_1, \Box\Pi^{\lambda_1} \Rightarrow \mid \Box\Gamma_2, \Box\Pi^{\lambda_2} \Rightarrow \Box C$ as required. $\Box$

Concluding Remark. In this paper, we have introduced sequent of relations and hypersequent calculi and established PSPACE completeness for a basic Gödel modal logic. Our broader goal is to extend these results to classes of modal fuzzy logics. Certain Gödel modal logics with accessibility relations that are transitive, reflexive, symmetric, etc. (giving e.g. $\mathsf{K4}$ and $\mathsf{S4}$ versions of the logic) are axiomatized in [6] and in these cases, it is a straightforward task to develop corresponding hypersequent calculi extending $\mathsf{GGK}_{\Box}$ that admit cut-elimination. However, a general treatment incorporating more complicated modal conditions

will require more expressive formalisms such as display calculi or labelled sequents/tableaux as is already the case for classical modal logics. Another task, important in connection with fuzzy description logics, is to develop calculi to deal with logics equipped also with the modality $\Diamond$. Although similar methods to those developed for the single modality should work, the logics based on standard and fuzzy Kripke frames diverge and require different rules.

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